Yifeng Huang

Postdoctoral Research/Teaching Fellow, Emory University
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Positions

2026–present Postdoctoral Research/Teaching Fellow, Emory University.
2024–2026 Assistant Professor (RTPC), University of Southern California.
2022–2024 Postdoctoral Research Fellow in Mathematics, University of British Columbia.

Education

2022 Ph.D. in Mathematics, University of Michigan. Advisor: Michael Zieve; co-advisor: Jeffrey Lagarias. Thesis: *Topics on Polynomial Equations in Noncommutative Rings and Motivic Aspects of Moduli Spaces.*
2015 B.Sc. in Mathematics, First Class Honors, Hong Kong University of Science and Technology.

Research Interests

Algebraic geometry and number theory, especially their combinatorial aspects; zero-dimensional enumerative geometry, q-series, discrete random matrix theory, and Diophantine equations on noncommutative algebras.

Publications

[1] Yifeng Huang and Ruofan Jiang. "Flag Quot and Coh zeta functions". *Contemp. Math.*, accepted.
[2] Yifeng Huang. "Commuting matrices via commuting endomorphisms". *Linear Algebra Appl.*, accepted. arXiv:2404.19483.
[3] Yifeng Huang. "A quadratic form generalization of rational `dinv`". *Res. Math. Sci.* 13 (2026), Article 44.
[4] Yifeng Huang, Borys Kadets, and Olivier Martin. "Low degree subvarieties of universal hypersurfaces". *J. Reine Angew. Math.* 833 (2026), 317–337.
[5] Jason Bell, Dragos Ghioca, and Yifeng Huang. "A Mordell–Lang-type problem for GL_m". *Bull. Aust. Math. Soc.* 111(3) (2025), 433–444.
[6] Yifeng Huang. "Cokernels of random matrix products and flag Cohen–Lenstra heuristic". *Forum Math.* 37(5) (2025), 1325–1332.
[7] Dragos Ghioca and Yifeng Huang. "A non-abelian variant of the classical Mordell–Lang conjecture". *Manuscripta Math.* 176 (2025), Paper No. 71.
[8] Yifeng Huang and Ruofan Jiang. "Lattices in $\mathbb{F}_q[[T]]^d$ and spiral shifting operators". *Adv. in Appl. Math.* 171 (2025), 102950.
[9] Gilyoung Cheong and Yifeng Huang. "The cokernel of a polynomial push-forward of a random integral matrix with concentrated residue". *Math. Proc. Cambridge Philos. Soc.* 178(2) (2025), 229–257.
[10] Yifeng Huang, Ken Ono, and Hasan Saad. "Counting matrix points on certain varieties over finite fields". *Contemp. Math.* 818 (2025), 181–195.
[11] Alexander Clifton, Bishal Deb, Yifeng Huang, Sam Spiro, and Semin Yoo. "Continuously increasing subsequences of random multiset permutations". *European J. Combin.* 110 (2023), 103708.
[12] Yifeng Huang. "Mutually annihilating matrices, and a Cohen–Lenstra series for the nodal singularity". *J. Algebra* 619 (2023), 26–50.
[13] Yifeng Huang. "Counting on the variety of modules over the quantum plane". *Algebr. Comb.* 5(3) (2022), 583–592.
[14] Gilyoung Cheong and Yifeng Huang. "Betti and Hodge numbers of configuration spaces of a punctured elliptic curve from its zeta functions". *Trans. Amer. Math. Soc.* 375(9) (2022), 6363–6383.
[15] Gilyoung Cheong and Yifeng Huang. "Cohen–Lenstra distributions via random matrices over complete discrete valuation rings with finite residue fields". *Illinois J. Math.* 65(2) (2021), 385–415.
[16] Yifeng Huang. "Unit equations on quaternions". *Q. J. Math.* 71(4) (2020), 1521–1534.

Preprints

[1] Yifeng Huang, Kenny Lau, and Ken Ono. "Rogers–Ramanujan identities from the geometry of $X^a = Y^b$". Submitted, 2026. arXiv:2609.20567.
[2] Yifeng Huang, Ruofan Jiang, and Alexei Oblomkov. "Quot scheme of points on torus knot singularities". 2026. arXiv:2608.16086.
[3] Yifeng Huang, Kenny Lau, Ken Ono, and Peter Paule. "Algebraic geometric framework of Rogers–Ramanujan identities". Submitted, 2026. arXiv:2608.15219.
[4] Shane Chern and Yifeng Huang. "Multiple Rogers–Ramanujan type identities for inert quadratic orders". Submitted. arXiv:2511.09452.
[5] Asvin G, Yifeng Huang, Ruofan Jiang, and Yifan Wei. "Matrix points on varieties". Submitted. arXiv:2510.13380.
[6] Yifeng Huang, Hoi H. Nguyen, and Roger Van Peski. "Cohen–Lenstra flag universality for random matrix products". Submitted. arXiv:2508.10127.
[7] Yifeng Huang. "Coh zeta functions for inert quadratic orders". Submitted. arXiv:2507.21966.
[8] Yifeng Huang and Eric Ramos. "Hilbert series for configuration spaces of punctured surfaces". Submitted. arXiv:2507.09746.

[9] Yifeng Huang and Ruofan Jiang. "Motivic Coh and Quot zeta functions of singular curves". Submitted. arXiv:2312.12528.
[10] Yifeng Huang and Ruofan Jiang. "Punctual Quot schemes and Cohen–Lenstra series of the cusp singularity". arXiv:2305.06411.
[11] Yifeng Huang. "Cohomology of configuration spaces on punctured varieties". Submitted. arXiv:2011.07153.
[12] Tianyu Wang, Yifeng Huang, and Didong Li. "From the Greene–Wu Convolution to Gradient Estimation over Riemannian Manifolds". Submitted. arXiv:2108.07406.

Grants and Awards

2024 UBC Teaching Award, recognized by the Faculty of Science for the highest Student Experience of Instruction scores in Winter 2024 for MATH 200 (Calculus III).

2022 AMS–Simons Travel Grant.

2019 University of Michigan Mathematics Department Summer Research Grant, funded by the Indu and Gopal Prasad Family Fund.

Teaching

2026 Emory, MATH 221 (Linear Algebra), lecturer, two classes of 40.

2025 USC, MATH 430 (Theory of Numbers), lecturer, class of 30; MATH 118 (Calculus for Business and Economics), lecturer, class of 60.

2024–2025 USC, MATH 432 (Applied Combinatorics), lecturer, class of 30.

2024 USC, MATH 225 (Linear Algebra and Differential Equations), lecturer, class of 50; UBC, MATH 200 (Calculus III), lecturer, class of 200.

2023 UBC, MATH 221 (Matrix Algebra), lecturer, class of 90; MATH 101 (Integral Calculus), instructor of three interactive classes of 60.

2022 UBC, MATH 100 (Differential Calculus), instructor of four interactive classes of 60.

2020 University of Michigan, EECS 203 (Discrete Mathematics), lecturer, class of 200.

2017–2019 University of Michigan, MATH 116 (Calculus II), instructor of an interactive class of 20.

2018 University of Michigan, MATH 676 (Algebraic Number Theory), grader.

2016, 2021 University of Michigan, MATH 115 (Calculus I), instructor of an interactive class of 20.

2015 University of Michigan, MATH 105 (Precalculus), instructor of an interactive class of 20.

Mentoring

2026 Caltech SURF, Postdoctoral Mentor.

Spring 2025 USC, MATH 490 (Directed Reading), mentored student research.

2024–2025 Harvey Mudd College, co-advisor of undergraduate thesis.

2023 University of Virginia, research mentor, REU in Number Theory led by Ken Ono.

Invited Talks

2026 Kennesaw State University, AMS Fall Southeastern Sectional, Special Session on Representation Theory with Applications in Geometry, Algebra and Mathematical Physics.

2026 Kennesaw State University, AMS Fall Southeastern Sectional, Special Session on Connections between q-Series, Partitions and Modular Forms.

2026 Louisiana State University, Algebra & Number Theory Seminar.

2026 Cleveland, Ohio, SIAM Annual Meeting (AN26), Minisymposium "Special Functions with Applications in Number Theory and Combinatorics".

2026 California State University, Northridge, Mathematics Colloquium.

2026 Ohio State University, Mini-course on Cohen–Lenstra-type heuristics for random matrices (1-hour lecture).

2026 Boston College, AMS Spring Eastern Sectional, Special Session "Configuration Spaces and Applications" (45-minute talk).

2025 AMS Fall Virtual Eastern Sectional, Special Session "Representations of p-adic Groups and Noncommutative Geometry" (50-minute talk).

2025 UC Irvine, Number Theory Seminar.

2025 Montréal, Third Joint SIAM/CAIMS Annual Meetings, "Hypergeometric Series and Their Applications" (25-minute talk).

2025 HKUST, Algebra and Geometry Seminar.

2025 UCSD, Algebra Seminar.
2025 Seattle, Joint Mathematics Meeting, Special Session "Quaternions" (20-minute talk).
2024 UMass Amherst, Valley Geometry Seminar.
2023 UCSD, Combinatorics Seminar.
2023 Southern California Number Theory Day.
2023 UCSD, Algebraic Geometry Seminar.
2023 Simon Fraser University, Number Theory and Algebraic Geometry Seminar.
2023 Boston, Joint Mathematics Meeting, Special Session "Quaternions".
2023 Boston, Joint Mathematics Meeting, Special Session "Combinatorics and Geometry of Jordan Type and Lefschetz Properties".
2022–2023 Virginia Tech, Algebra Seminar (two talks).
2022 University of Virginia, Algebra Seminar.
2022 UMass Amherst, AMS Eastern Sectional, Special Session "Young Voices in Combinatorics".
2022 UC Irvine, Number Theory Seminar.
2022 USC, Combinatorics Seminar.
2022 Rutgers, Graduate Algebra and Representation Theory Seminar.
2021 St. John's University, NYC Noncommutative Geometry Seminar.
2021 University of Michigan, RTG Seminar on Number Theory.
2020 UBC, Discrete Mathematics Seminar.
2020 UBC, Algebraic Geometry Seminar.
2020 University of Waterloo, Algebra Seminar.
2020 Rutgers, Algebra Seminar.
2020 University of Minnesota, Combinatorics and Commutative Algebra Seminar.

Talks in Local Seminars

2024–2026 USC, Combinatorics Seminar (three talks).
2025 USC, Topology Seminar.
2025 Caltech, USC/Caltech Joint Algebra and Geometry Seminar.
2025 USC, Algebra Seminar.
2025 Caltech, Number Theory Seminar.
2023 UBC, Algebraic Geometry Seminar (two talks).
2022–2023 UBC, Number Theory Seminar (two talks).
2023 UBC, Discrete Mathematics Seminar.
2021 University of Michigan, RTG Seminar on Number Theory.
2020 University of Michigan, RTG Seminar on Number Theory.

Organization and Service

2026 SIAM Annual Meeting (AN26), organizer of the minisymposium "Hypergeometric Series and Modular Forms".
2024–2025 Organizer, USC/Caltech Joint Algebra and Geometry Seminar.
2022–2024 Organizer, UBC Algebraic Geometry Seminar.

Reviewing

Reviewer, MathSciNet.
Referee for *American Journal of Mathematics*, *Annals of Probability*, *Electronic Communications in Probability*, *Forum Mathematicum*, *Advances in Mathematics*, *Linear Algebra and its Applications*, *Linear and Multilinear Algebra*, *Journal of Algebra*, *Journal of Symbolic Computation*, *Combinatorics, Probability and Computing*, and *Indagationes Mathematicae*.

Outreach

2024–2025 UCLA Olga Radko Endowed Math Circle, Advanced III lead instructor.
2023 Virginia Tech Blacksburg Math Circle, one-hour lecture on Pólya enumeration theorem to grades 4–8.

Skills

Languages English (fluent), Mandarin Chinese (native), Cantonese (native), Spanish (reading), French (reading).

Coding C++, Python, Mathematica, Sage.